

Lab 2: Empirical Risk and Simple Linear Regression **Solutions**

EECS 245, Fall 2026 at the University of Michigan

due by the end of your lab section on Wednesday, September 9th, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, some of which will involve writing code and others that will involve writing math on paper. To receive credit for a lab, you must complete all required activities and show your lab TA by the end of the lab section.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Recap: The Modeling Recipe

In [Chapter 1.3](#), we introduced the three-step modeling recipe for finding optimal model parameters, which ultimately helps us make the best possible predictions.

1. Choose a model.

$$\underbrace{h(x_i) = w}_{\text{constant model}} \quad \underbrace{h(x_i) = w_0 + w_1 x_i}_{\text{simple linear regression model}}$$

2. Choose a loss function.

$$\underbrace{L_{\text{sq}}(y_i, h(x_i)) = (y_i - h(x_i))^2}_{\text{squared loss}} \quad \underbrace{L_{\text{abs}}(y_i, h(x_i)) = |y_i - h(x_i)|}_{\text{absolute loss}}$$

3. Minimize *average loss* (also called *empirical risk*) to find optimal model parameters.

- Constant model, squared loss: $R_{\text{sq}}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2 \implies w^* = \bar{y}$
- Constant model, absolute loss: $R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w| \implies w^* = \text{Median}(y_1, y_2, \dots, y_n)$
- Simple linear regression model, squared loss:

$$R_{\text{sq}}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2 \implies w_1^* = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad w_0^* = \bar{y} - w_1^* \bar{x}$$

Activity 1: Relative Squared Loss

Suppose we'd like to find the optimal parameter, w^* , for the constant model $h(x_i) = w$. To do so, we use the following loss function, called the **relative squared loss**:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

What value of w minimizes the average loss (i.e. empirical risk) when using the relative squared loss function – that is, what is w^* ? Your answer should only be in terms of the variables $n, y_1, y_2, \dots, y_n$, and any constants.

Solution:

Since $h(x_i) = w$ for the constant model, relative squared loss for the constant model is:

$$L_{\text{rsq}}(y_i, w) = \frac{(y_i - w)^2}{y_i}$$

and so average relative squared loss for the constant model is:

$$R_{\text{rsq}}(w) = \frac{1}{n} \sum_{i=1}^n \frac{(y_i - w)^2}{y_i}$$

To find the value of w that minimizes $R_{\text{rsq}}(w)$, we'll first find its first derivative and set it to zero. The first derivative of $R_{\text{rsq}}(w)$ is:

$$\begin{aligned} \frac{d}{dw} R_{\text{rsq}}(w) &= \frac{d}{dw} \left(\frac{1}{n} \sum_{i=1}^n \frac{(y_i - w)^2}{y_i} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{d}{dw} \left(\frac{(y_i - w)^2}{y_i} \right) \end{aligned}$$

At this point, it'll be useful to step aside and find the derivative of $L_{\text{rsq}}(y_i, w)$ with respect to w , as this is the expression being summed. The derivative of $L_{\text{rsq}}(y_i, w)$ with respect to w is:

$$\begin{aligned} \frac{d}{dw} L_{\text{rsq}}(y_i, w) &= \frac{d}{dw} \frac{(y_i - w)^2}{y_i} \\ &= \frac{1}{y_i} \cdot \frac{d}{dw} (y_i - w)^2 \\ &= \frac{1}{y_i} \cdot 2(y_i - w) \cdot (-1) \\ &= -2 \cdot \frac{y_i - w}{y_i} \\ &= \boxed{2 \cdot \frac{w}{y_i} - 2} \end{aligned}$$

Back to $\frac{d}{dw} R_{\text{rsq}}(w)$, we have:

$$\begin{aligned} \frac{d}{dw} R_{\text{rsq}}(w) &= \frac{1}{n} \sum_{i=1}^n \frac{d}{dw} \left(\frac{(y_i - w)^2}{y_i} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(2 \cdot \frac{w}{y_i} - 2 \right) \\ &= \frac{2w}{n} \sum_{i=1}^n \left(\frac{1}{y_i} \right) - \frac{1}{n} \sum_{i=1}^n 2 \\ &= \frac{2w}{n} \sum_{i=1}^n \left(\frac{1}{y_i} \right) - 2 \end{aligned}$$

Setting this equal to 0 yields:

$$\frac{2w}{n} \sum_{i=1}^n \left(\frac{1}{y_i}\right) - 2 = 0$$

$$\frac{w}{n} \sum_{i=1}^n \left(\frac{1}{y_i}\right) = 1$$

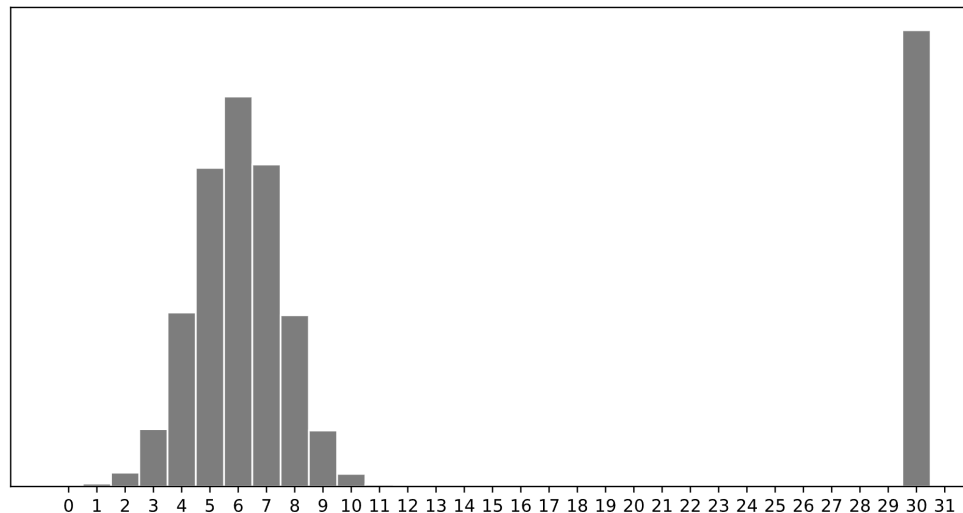
$$w^* = \frac{1}{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{y_i}\right)}$$

$$w^* = \boxed{\frac{n}{\sum_{i=1}^n \frac{1}{y_i}}}$$

This is known as the **harmonic mean** of $y_1, y_2, \dots, y_n$.

Activity 2: Rapid Fire

Consider a dataset of n **integers**, $y_1, y_2, \dots, y_n$, whose histogram is given below:



a) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n \begin{cases} 0 & y_i = w \\ 1 & y_i \neq w \end{cases}$$

- ☐ 1
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 11
 ☐ 15
 ☐ 30

Solution: 30.

The minimizer of average 0-1 loss is the **mode**.

See: [Chapter 1.4: Beyond Absolute and Squared Loss](#)

b) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n |y_i - w|$$

- ☐ 1
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 11
 ☐ 15
 ☐ 30

Solution: 7.

The minimizer of average absolute loss is the **median**. The outliers near 30 shift it from 6 to 7.

c) Which of the following is closest to the constant prediction w^* that minimizes:

$$\frac{1}{n} \sum_{i=1}^n (y_i - w)^2$$

- ☐ 1
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 11
 ☐ 15
 ☐ 30

Solution: 11.

The minimizer of average squared loss is the **mean**, pulled upward by the heavy right tail, so it's above the median (7) and closest to 11.

d) Which of the following is closest to the constant prediction w^* that minimizes:

$$\lim_{p \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

Hint: Think about the effect of outliers.

☐ 1 ☐ 5 ☐ 6 ☐ 7 ☐ 11 ☐ 15 ☐ 30

Solution: 15.

As $p \rightarrow \infty$, the minimizer is the **midrange**, halfway between min and max.

Activity 3: Slope of Mean Absolute Error

Consider a dataset of 8 points, $y_1, y_2, \dots, y_8$ that are in sorted order, i.e. $y_1 < y_2 < \dots < y_8$.

Recall that mean absolute error, $R_{\text{abs}}(w)$, for the constant model $h(x_i) = w$ is defined as:

$$R_{\text{abs}}(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|$$

This is a piecewise linear function that changes slope at each data point. The slope of $R_{\text{abs}}(w)$ at any w that is not a data point is:

$$\frac{d}{dw} R_{\text{abs}}(w) = \frac{\# \text{ of points left of } w - \# \text{ of points right of } w}{n}, \quad w \notin \{y_1, \dots, y_n\}$$

Suppose that $y_4 = 10$, $y_5 = 14$, $y_6 = 22$, and $R_{\text{abs}}(11) = 9$. What is $R_{\text{abs}}(22)$?

Hint: You don't have all 8 of the y -values, so you can't find $R_{\text{abs}}(22)$ just by plugging in numbers into the formula for $R_{\text{abs}}(w)$. Instead, think about how to use the slope formula.

Solution:

$$R_{\text{abs}}(22) = 11.$$

We can write the points given to us as:

$$y_1, y_2, y_3, 10, 14, 22, y_7, y_8$$

Since there are an even number of data points ($n = 8$), the minimizer of absolute error is not a single point but the entire interval between the two middle points. Here, the middle two are 10 and 14, so every $w \in [10, 14]$ minimizes $R_{\text{abs}}(w)$. This explains why the error is *flat* inside that interval: the number of points on the left equals the number on the right, so shifting w around does not change the error. As a result, $R_{\text{abs}}(11) = 9$ and $R_{\text{abs}}(14) = 9$.

Once we move beyond 14, the balance breaks. There are now five points to the left and only three to the right, so the slope of $R_{\text{abs}}(w)$ becomes positive. The slope formula tells us:

$$\frac{d}{dw} R_{\text{abs}}(w) = \frac{\# \text{ of points left of } w - \# \text{ of points right of } w}{n}, \quad w \notin \{y_1, \dots, y_n\}$$

so for any $w \in (14, 22)$ we have

$$\frac{d}{dw} R_{\text{abs}}(w) = \frac{5 - 3}{8} = \frac{1}{4}.$$

This means that for every one unit we move to the right of $w = 14$, the error increases by $\frac{1}{4}$. Moving from $w = 14$ to $w = 22$ is a distance of $22 - 14 = 8$ units, so the error increases by

$$8 \cdot \frac{1}{4} = 2.$$

Adding this to the baseline error of $R_{\text{abs}}(14) = 9$, we get:

$$\begin{aligned} R_{\text{abs}}(22) &= R_{\text{abs}}(14) + (22 - 14) \cdot \frac{1}{4} \\ &= 9 + 2 = 11. \end{aligned}$$

Activity 4: Programming

Complete the tasks in the lab02.ipynb notebook.

There are two ways to access the supplemental Jupyter Notebook:

- **Option 1 (preferred):** Set up a Jupyter Notebook environment locally, use git to clone our [course repository](#), and open labs/lab02/lab02.ipynb. For instructions on how to do this, see the [Environment Setup](#) page of the course website.
- **Option 2:** Click [here](#) to open lab02.ipynb on DataHub. Before doing so, read the instructions on the [Environment Setup](#) page on how to use the DataHub.

To receive credit for Activity 4, you'll need to show your lab TA that all test cases have passed.

Activity 5: What Do You Mean?

Suppose we want to fit a simple linear model (using squared loss) that predicts the number of ingredients in a product given its price. We're given that:

- The average cost of a product in our dataset is \$40, i.e. $\bar{x} = 40$
- The average number of ingredients in a product in our dataset is 15, i.e. $\bar{y} = 15$

The intercept and slope of the regression line are $w_0^* = 11$ and $w_1^* = \frac{1}{10}$, respectively.

- a) Suppose Victors' Veil (a skincare product) costs \$40 and has 11 ingredients. What is the squared loss of our model's predicted number of ingredients for Victors' Veil?

Solution: Using the equation of the regression model we have seen in class:

$$h(x_i) = w_0^* + w_1^* x_i$$

Plugging in $w_0^* = 11$, $w_1^* = \frac{1}{10}$, and $x = 40$ gives us:

$$h(x_i) = 11 + \frac{1}{10} \cdot 40 = 15$$

The squared loss is $L = (y_i - h(x_i))^2$, substituting $y = 11$ (actual) and $h(x_i) = 15$ (predicted) gives us:

$$L = (11 - 15)^2 = 16$$

- b) Is it possible to answer part a) above **just** by knowing $\bar{x}$ and $\bar{y}$, i.e. **without** knowing the values of w_0^* and w_1^* ? Once you select an answer, explain it to your peers.

Hint: If you're stuck, look through the headings of the sections of [Chapter 2.3](#).

- ☒ Yes, it's possible ☐ No, it's not possible

Solution: Yes, the values of w_0^* and w_1^* don't impact the answer to part a).

The simple linear model minimizing mean squared error will always go through the point $(\bar{x}, \bar{y})$. We're given $\bar{x} = 40$ and $\bar{y} = 15$, meaning that for a product that costs \$40 we will predict that it has 15 ingredients, no matter what the slope and intercept end up being.

Activity 6: The Meaning of Mean Squared Error

Suppose we'd like to predict the number of minutes a delivery will take, y , as a function of distance, x . To do so, we look to our dataset of n deliveries, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, and fit two simple linear models:

- $F(x_i) = a_0 + a_1 x_i$, where:

$$a_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

Here, $\bar{x}$ and $\bar{y}$ are the respective means of the x -values and y -values.

- $G(x_i) = b_0 + b_1 x_i$, where b_0 and b_1 are chosen such that $G(x_i) = b_0 + b_1 x_i$ minimizes **mean absolute error** on the dataset. Assume that no other line minimizes mean absolute error on the dataset, i.e. that the values of b_0 and b_1 are unique. Assume also that F and G are distinct lines.

a) Fill in the ???:

$$\sum_{i=1}^n (y_i - F(x_i))^2 \quad \text{???} \quad \sum_{i=1}^n (y_i - G(x_i))^2$$

☐ $>$ ☐ $\geq$ ☐ $=$ ☐ $\leq$ ☒ $<$ ☐ Impossible to tell

Solution: The quantity on the left hand side is the **total squared error** of model F . By definition, F , is the line that minimizes MSE over all possible linear models.

The quantity on the right hand side is the total squared error of model G . However, G is optimized for **mean absolute error**, not MSE.

The question tells us that F and G are different, so $\sum_{i=1}^n (y_i - F(x_i))^2 < \sum_{i=1}^n (y_i - G(x_i))^2$ must be true.

b) Fill in the ???:

$$\left(\sum_{i=1}^n |y_i - F(x_i)| \right)^2 \quad \text{???} \quad \left(\sum_{i=1}^n |y_i - G(x_i)| \right)^2$$

☒ $>$ ☐ $\geq$ ☐ $=$ ☐ $\leq$ ☐ $<$ ☐ Impossible to tell

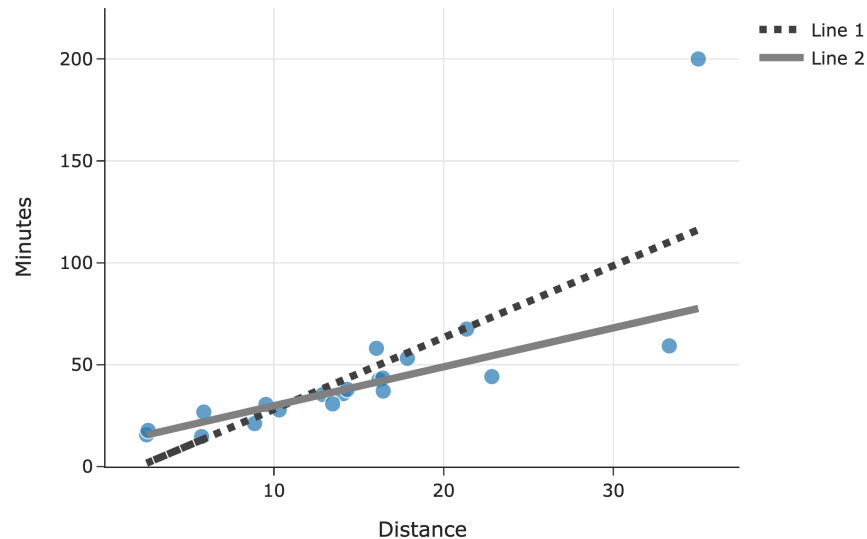
Solution: The quantity on the left hand side is the **squared total absolute error** of model F .

The quantity on the right hand side is the **squared total absolute error** of model G . By definition, G , is the line that minimizes MAE over all possible linear models.

The total absolute error of G must be less than the total absolute error of F , and squaring them doesn't change that relationship because both totals are guaranteed to be positive numbers (sums of absolute values).

Finally, F and G are different, so $\left(\sum_{i=1}^n |y_i - F(x_i)|\right)^2 > \left(\sum_{i=1}^n |y_i - G(x_i)|\right)^2$

- c) Below, we've drawn the lines for both F and G along with a scatter plot for the original n deliveries:



Which line corresponds to F ?

● Line 1

○ Line 2

Solution: Line 1

The key idea is that models trained with squared loss (MSE) are more sensitive to outliers than models trained with absolute loss (MAE).

Since Line 1 appears to be “pulled up” more strongly by an outlier, it suggests that this line was influenced more heavily by extreme values. This behavior aligns with how MSE-based regression works: outliers have a greater impact on the overall loss because squaring the errors makes large deviations even more significant.

In contrast, MAE-based regression (Line 2) is less sensitive to outliers because absolute differences do not grow as quickly.

Therefore, Line 1 corresponds to F , the MSE-minimizing line.

The following are extra practice. Don't feel pressured to answer all of these problems in lab, but make sure to attempt them at some point.

Activity 7: Relative Squared Loss, Continued

Recall the formula for **relative squared loss** from Activity 1:

$$L_{\text{rsq}}(y_i, h(x_i)) = \frac{(y_i - h(x_i))^2}{y_i}$$

- a) Let $C(y_1, y_2, \dots, y_n)$ be your minimizer w^* from Activity 1. That is, for a particular dataset $y_1, y_2, \dots, y_n$, $C(y_1, y_2, \dots, y_n)$ is the value of w that minimizes empirical risk for relative squared loss on that dataset.

What is the value of $\lim_{y_4 \rightarrow \infty} C(1, 3, 5, y_4)$ in terms of $C(1, 3, 5)$? Your answer should involve the function C and/or one or more constants.

Hint: To notice the pattern, evaluate $C(1, 3, 5, 100)$, $C(1, 3, 5, 10000)$, and $C(1, 3, 5, 1000000)$.

Solution:

$$\begin{aligned} \lim_{y_4 \rightarrow \infty} C(1, 3, 5, y_4) &= \lim_{y_4 \rightarrow \infty} \frac{4}{\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \frac{1}{y_4}} \\ &= \frac{4}{\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + 0} \\ &= \frac{4}{3} \cdot \frac{3}{\frac{1}{1} + \frac{1}{3} + \frac{1}{5}} \\ &= \frac{4}{3} \cdot C(1, 3, 5) \end{aligned}$$

- b) What is the value of $\lim_{y_4 \rightarrow 0} C(1, 3, 5, y_4)$? Again, your answer should involve the function C and/or one or more constants.

Solution:

$$\begin{aligned} \lim_{y_4 \rightarrow 0} C(1, 3, 5, y_4) &= \lim_{y_4 \rightarrow 0} \frac{4}{\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \frac{1}{y_4}} \\ &= \frac{4}{\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \infty} \\ &= \frac{4}{\infty} = 0 \end{aligned}$$

- c) Based on the results of the previous two parts, when is the prediction $C(y_1, y_2, \dots, y_n)$ robust to outliers? When is it not robust to outliers?

Solution:

$C(y_1, y_2, \dots, y_n)$ is great at ignoring large outliers. No matter how large you make any particular value, $C(y_1, y_2, \dots, y_n)$ is upper-bounded by $\frac{n}{n-1}$ multiplied by the value of C applied to all data points excluding the large outlier. This is as opposed to the regular “arithmetic mean”, where if you make a single data point arbitrarily large, the mean also becomes arbitrarily large (i.e. if $y_n \rightarrow \infty$, then $\text{Mean}(y_1, y_2, \dots, y_n) \rightarrow \infty$ too).

However, $C(y_1, y_2, \dots, y_n)$ is not robust to small outliers. As a particular data point approaches 0, the value of $C(y_1, y_2, \dots, y_n)$ also approaches 0 no matter how large the other data points are.

Activity 8: A Refresher

Consider a dataset of $y_1, y_2, \dots, y_n$, all of which are **positive**. We want to fit a constant model, $h(x_i) = w$, to the data.

Let w_p^* be the optimal constant prediction that minimizes average degree- p loss, $R_p(w)$, defined below:

$$R_p(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^p$$

For example, w_2^* is the optimal constant prediction that minimizes $R_2(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^2$

a) In each of the parts below, determine the value of the quantity provided. By “the data”, we are referring to $y_1, y_2, \dots, y_n$. The answer choices are as follows; **select one item in each row**.

A: The standard deviation of the data

B: The variance of the data

C: The mean of the data

D: The median of the data

E: The midrange of the data, $\frac{y_{\min} + y_{\max}}{2}$

F: The mode of the data

G: None of these

	A	B	C	D	E	F	G
(i) w_0^*	☐	☐	☐	☐	☐	☐	☒
(ii) w_1^*	☐	☐	☐	☒	☐	☐	☐
(iii) $R_1(w_1^*)$	☐	☐	☐	☐	☐	☐	☒
(iv) w_2^*	☐	☐	☒	☐	☐	☐	☐
(v) $R_2(w_2^*)$	☐	☒	☐	☐	☐	☐	☐

Solution:

- (i) **None of these.** When $w = y_i$, the term $|y_i - w|^0$ becomes 0^0 , which is undefined. Thus, $R_0(w)$ is not well-defined for all w as written. The intended loss function was 0-1 loss, which would have made the empirical risk

$$R_{0,1}(w) = \frac{1}{n} \sum_{i=1}^n L_{0,1}(y_i, w) = \frac{\text{number of points not equal to } w}{n}.$$

To minimize this empirical risk, we would choose w to be the **mode** of the data, since it is the value that the greatest number of data points are equal to.

- (ii) w_1^* is the median of the data, since $R_1(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|$
- (iii) $R_1(w_1^*)$ is the minimum mean absolute error, which is none of these.
- (iv) w_2^* is the mean of the data, since $R_2(w) = \frac{1}{n} \sum_{i=1}^n |y_i - w|^2$ is equivalent to mean squared error.
- (v) $R_2(w_2^*)$ is the variance of the data, or the minimum mean squared error, shown below:

$$\begin{aligned} R_2(w_2^*) &= \frac{1}{n} \sum_{i=1}^n |y_i - w_2^*|^2 \\ &= \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}|^2 \\ &= \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 = \sigma_y^2 \end{aligned}$$

- b) Now, suppose we want to find the optimal constant prediction, w_U^* , using the "Ultra" loss function, defined below:

$$L_U(y_i, w) = y_i(y_i - w)^2$$

To find w_U^* , we minimize $R_U(w)$, the average Ultra loss. How does w_U^* compare to the mean of the data, M ?

- ☐ $w_U^* > M$ ☒ $w_U^* \geq M$ ☐ $w_U^* = M$ ☐ $w_U^* \leq M$ ☐ $w_U^* < M$

Solution: Minimizing the average Ultra loss means minimizing the empirical risk:

$$R_U(w) = \frac{1}{n} \sum_{i=1}^n y_i(y_i - w)^2$$

The minimizer is $w_U^* = \frac{\sum_{i=1}^n y_i^2}{\sum_{i=1}^n y_i}$. Since $M > 0$, we have $w_U^* - M = \frac{\frac{1}{n} \sum_{i=1}^n (y_i - M)^2}{M} \geq 0$. Thus, $w_U^* \geq M$, with equality exactly when all the y_i values are equal.

- c) Finally, to find the optimal constant prediction, we will instead minimize **regularized** average Ultra loss, $R_\lambda(w)$, defined below:

$$R_\lambda(w) = \left(\frac{1}{n} \sum_{i=1}^n y_i (y_i - w)^2 \right) + \lambda w^2$$

Here, assume $\lambda > 0$ is some positive constant. (We will cover regularization in more detail later in the term.)

Find w^* , the constant prediction that minimizes $R_\lambda(w)$. Give your answer as an expression in terms of the y_i 's, n , and/or λ .

Solution: To minimize the regularized average Ultra loss, we solve for w by setting $\frac{\partial R}{\partial w} = 0$ and solving for w .

Step 1: Compute the derivative and set to 0.

$$\begin{aligned}\frac{\partial R_\lambda}{\partial w}(w) &= \frac{\partial}{\partial w} \left[\left(\frac{1}{n} \sum_{i=1}^n y_i (y_i - w)^2 \right) + \lambda w^2 \right] \\ &= -2 \left(\frac{1}{n} \sum_{i=1}^n y_i (y_i - w) \right) + 2\lambda w = 0\end{aligned}$$

Step 2: Expand and simplify.

$$\begin{aligned}-2 \left(\frac{1}{n} \sum_{i=1}^n y_i (y_i - w) \right) + 2\lambda w &= 0 \\ \frac{1}{n} \sum_{i=1}^n y_i (y_i - w) - \lambda w &= 0 \\ \frac{1}{n} \sum_{i=1}^n y_i^2 - \frac{1}{n} \sum_{i=1}^n y_i w - \lambda w &= 0 \\ \frac{1}{n} \sum_{i=1}^n y_i^2 &= \frac{1}{n} \sum_{i=1}^n y_i w + \lambda w\end{aligned}$$

Step 3: Solve for w .

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n y_i^2 &= \frac{1}{n} \sum_{i=1}^n y_i w + \lambda w \\ \frac{1}{n} \sum_{i=1}^n y_i^2 &= w \left(\frac{1}{n} \sum_{i=1}^n y_i + \lambda \right) \\ \frac{\frac{1}{n} \sum_{i=1}^n y_i^2}{\frac{1}{n} \sum_{i=1}^n y_i + \lambda} &= w\end{aligned}$$

Step 4: Multiply by $\frac{n}{n}$

$$w^* = \frac{\sum_{i=1}^n y_i^2}{\sum_{i=1}^n y_i + n\lambda}$$